Day 14 - Notes

Flow Logic, Approval Workflows & Enterprise Governance Notes

Flow Builder Logic

What is Flow Builder?

Flow Builder is Salesforce's automation tool used to automate business processes without writing traditional code.

It helps organizations automate:

- Data updates
- Notifications
- Record creation
- Approval routing
- Decision-based actions
- User interactions
- Complex workflow execution

Flow Builder supports enterprise automation by allowing organizations to build scalable and structured business logic.

Core Components of Flow Logic

Decision Elements

What is a Decision Element?

A Decision element controls branching inside a flow.

It checks conditions and sends the flow through different paths depending on the result.

Purpose of Decision Elements

Decision elements allow automation to behave differently based on:

- Record values
- User input
- Permissions
- Business rules
- Status conditions

Example

Attendance-based automation:

- If attendance < 75%
  → Send warning email
- If attendance < 60%
  → Notify parents
- If attendance < 50%
  → Escalate to administration

Importance of Decision Elements

Decision nodes are important because they:

- Control business logic
- Enable dynamic automation
- Prevent unnecessary actions
- Improve workflow flexibility
- Support enterprise rules and policies

Without decision logic, all users and records would follow the same process regardless of conditions.

Branching Workflows

What are Branching Workflows?

Branching workflows divide automation into multiple paths based on conditions.

Each branch performs different actions.

Benefits of Branching Logic

- Supports multiple scenarios
- Handles exceptions
- Improves automation intelligence
- Reduces manual intervention
- Creates structured business processes

Enterprise Example

A scholarship approval workflow may branch depending on:

- Student marks
- Family income
- Attendance percentage
- Scholarship type

Each condition triggers different approval paths.

Variables in Flows

What are Variables?

Variables temporarily store data inside flows.

They act as containers that hold values during automation execution.

Types of Data Stored

Variables can store:

- Text
- Numbers
- Dates
- Currency
- Record values
- User input

Why Variables are Important

Variables help flows:

- Pass information between elements
- Store intermediate values
- Reuse data
- Update records dynamically
- Support calculations

Example

A flow may store:

- Opportunity amount
- Discount percentage
- Account ID
- User input
- Approval status

inside variables for later processing.

Assignment Elements

What is an Assignment Element?

The Assignment element changes the value of variables during flow execution.

Assignment Operations

Assignment elements can:

- Replace values
- Add values
- Subtract values
- Append text
- Perform calculations

Example

Adding text:

Description = "VIP. " + Existing Description

Importance

Assignment elements help flows manipulate data dynamically before saving it to records.

Formula Logic in Flows

What are Flow Formulas?

Flow formulas automatically calculate values using expressions and functions.

Uses of Formulas

Formulas help:

- Convert data types
- Concatenate text
- Perform calculations
- Calculate dates
- Generate dynamic values

Example Formula

"Shipping Address added on " & TEXT(CurrentDate)

Why Formulas Matter

Formulas improve automation flexibility by creating values dynamically during runtime.

Multi-Step Workflows

What are Multi-Step Workflows?

Multi-step workflows are workflows containing several connected automation stages.

Features

They include:

- Sequential actions
- Conditional routing
- Approval chains
- Data updates
- Notifications
- User interaction

Enterprise Example

Course approval process:

1. Faculty submits course
2. Department head reviews
3. Academic committee approves
4. Administration finalizes
5. Course becomes active

Importance

Multi-step workflows ensure:

- Controlled execution
- Proper validation
- Business compliance
- Structured operations

Approval Processes

What is an Approval Process?

An approval process automates how records are reviewed and approved inside Salesforce.

Main Purpose

Approval processes ensure sensitive business actions are reviewed before becoming final.

Core Components

Approval processes include:

- Submission actions
- Approval steps
- Approvers
- Rejection handling
- Final approval actions
- Record locking

Approval Workflow Lifecycle

Step 1 — Record Submission

A user submits a record for approval.

Example:

- Discounted opportunity
- Leave request
- Scholarship request
- Budget request

Initial Actions

Common initial actions:

- Lock the record
- Send email notifications
- Update status fields
- Create tasks

Step 2 — Approval Routing

The system sends approval requests to designated approvers.

Approvers may include:

- Managers
- Team leads
- Department heads
- Finance teams
- Executives

Step 3 — Approval Decision

Approvers either:

- Approve
- Reject

based on business policies.

Step 4 — Final Actions

If Approved

Possible actions:

- Update status
- Unlock records
- Send confirmation notifications
- Trigger additional automation

If Rejected

Possible actions:

- Update rejection status
- Notify submitter
- Reopen requests
- Stop further processing

Multi-Level Approval Systems

What are Multi-Level Approvals?

Multi-level approvals require multiple approvers in sequence.

Example

Budget approval process:

1. Department Head
2. Finance Manager
3. Director
4. Executive Team

Why Multi-Level Approvals Matter

They ensure:

- Financial control
- Organizational accountability
- Risk reduction
- Policy enforcement

Governance in Enterprise Systems

What is Governance?

Governance refers to rules and controls that ensure systems operate safely, securely, and consistently.

Why Governance is Important

Enterprise systems handle:

- Sensitive business data
- Financial records
- Student records
- Employee information

- Customer operations
  Without governance, organizations face major risks.

Risks Without Governance

Security Risks

Unauthorized users may:

- Access sensitive records
- Modify protected data
- Bypass approval systems

Data Integrity Problems

Incorrect updates may:

- Corrupt records
- Cause inconsistent data
- Break automation logic

Business Risks

Improper changes may lead to:

- Financial loss
- Compliance violations
- Operational failures
- Reputation damage

Misuse of Permissions

Without restrictions:

- Anyone could approve records
- Critical workflows could be bypassed
- Fraud risks increase

Why Enterprises Restrict Sensitive Operations

Large organizations restrict important actions because systems affect thousands of users and business operations.

Restricted operations may include:

- Financial approvals
- Record deletion
- User access management
- Contract approval
- Budget modification

Benefits of Restrictions

- Improves security
- Reduces mistakes
- Maintains accountability
- Supports auditing
- Protects business operations

Custom Permissions in Flows

What are Custom Permissions?

Custom permissions control which users can access certain automation paths.

Example

Only onboarding managers may access project kickoff workflows.

Benefits

- Role-based automation
- Better security
- Controlled process execution
- User-specific workflow routing

Subflows

What is a Subflow?

A subflow is a flow that runs inside another flow.

Parent Flow

The main flow.

Child Flow

The reusable flow called by the parent flow.

Benefits of Subflows

Subflows help:

- Reduce duplicate automation
- Simplify maintenance
- Improve reusability
- Centralize business logic
- Improve scalability

Enterprise Advantage

Instead of updating the same logic in multiple flows, organizations update only one reusable child flow.

Enterprise Workflow Thinking

Why Enterprise Systems Need Controlled Workflows

Enterprise systems cannot allow unrestricted actions because organizations require:

- Security
- Validation
- Accountability
- Consistency
- Compliance
- Risk management
  Controlled workflows ensure that every action follows business policies.

Why Automation Must Follow Business Rules

Automation without business rules can:

- Approve incorrect records
- Create invalid data
- Trigger wrong actions
- Cause compliance violations
  Business rules ensure automation behaves correctly and safely.

Importance of Auditable Workflows

What is Auditability?

Auditability means organizations can track:

- Who performed actions
- When actions occurred
- What changes were made
- Why approvals happened

Why Auditability Matters

Auditable systems support:

- Compliance
- Security investigations
- Accountability
- Risk management
- Enterprise transparency

Enterprise Workflow Design Principles

Good Enterprise Workflow Design Includes

- Structured approvals
- Controlled access
- Decision-based routing
- Multi-step validation
- Error prevention
- Security restrictions
- Clear accountability
- Scalable automation

Key Concepts Learned

- Flow Builder automates business processes
- Decision elements create branching logic
- Variables store temporary data
- Assignment elements modify variable values
- Formulas calculate dynamic values
- Multi-step workflows structure enterprise operations
- Approval processes automate governance
- Multi-level approvals improve accountability
- Governance protects enterprise systems
- Custom permissions control workflow access
- Subflows improve reusability and scalability
- Enterprise automation must follow business rules

- Controlled workflows reduce organizational risk

Final Understanding

Day 14 provided understanding of how enterprise organizations build structured, secure, and governed automation systems using Salesforce Flows and Approval Processes.

The learning emphasized that enterprise automation is not just about speeding up tasks—it is about ensuring controlled operations, secure approvals, proper governance, accountability, and business rule enforcement across large-scale systems used by many users.